

A Neural Network Model For (Fashion-MNIST) Image Classification

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Overview

Implemented ANN (MLP) and CNN models in TensorFlow and Keras to classify images of Dataset Fashion-MNIST (60,000 training and 10,000 grayscale test images, 28×28 pixels, 10 categories).

- **ANN (MLP):** A simple model with a Flatten layer, two Dense hidden layers (128 & 64 units, ReLU), and Dropout (0.2) to prevent overfitting. Total parameters: 109,386.
- **CNN:** A more complex model with two Conv2D layers (32 & 64 filters) followed by MaxPooling, Dropout (0.3), a Flatten layer, and a final Dense layer (128 units, ReLU) with 0.5 Dropout, softmax. Total parameters: 225,034.

Preprocessing: Before training, I normalized all pixel values from 0-255 down to a 0-1 range. I also added a channel dimension for the CNN and one-hot encoded the 10 category labels. I held back 10% of the training data for validation.

Major Findings

- **Performance:** Test accuracy **ANN 88.25%** and for **CNN 90.85%**, (**+2.6% gain**) with comparable time (45-60 seconds on GPU). I trained both models for 17 epochs using the Adam optimizer and cross-entropy loss. CNN performs best in feature extraction.
- **Metrics:** Both models struggled to distinguish visually similar items like 'Shirt' and 'Coat'. The ANN had 180 misclassifications between 'Shirt' and 'Coat', while the CNN performed better with 140. The CNN's F1-scores highlighted this challenge, with a low of 0.72 for 'Shirt' but a high of 0.99 for 'Trouser', resulting in a 0.91 macro average.

Notable Innovations

- **Data Augmentation:** ImageDataGenerator for rotations/shifts/flips/zoom to boost variety and reduce overfitting (potential +1-2% accuracy if re-trained).
- **Regularization:** Dropout layers (0.2-0.5) minimized overfitting.
- **Visualization:** Curves/heatmaps for class-wise analysis.

Conclusions and Insights

The **CNN outperformed the ANN** because of its ability to process spatial hierarchies, meeting all project outcomes. Key insights were that **visual confusion** (e.g., Shirt/Coat) highlights the dataset's challenges and that **preprocessing** is key to success. **Future Improvements** to exceed **92% accuracy** would be by adding **batch normalization** or a **learning rate scheduler**.